# The maximal gain of nonadaptive randomization

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**Abstract**

We determine the supremum of the improvement in polynomial convergence rate due to randomization for approximation of bounded linear operators between real Banach spaces using nonadaptive continuous linear measurements. The supremum is one. We construct fixed infinite-dimensional operators whose gains tend to one, with worst-case expected-norm error and at most $n$ measurements for every random seed. The construction combines a finite-dimensional deterministic lower bound with a known nonadaptive CountSketch estimate and a weighted $\ell_1$ sum of blocks.

## 1 Introduction

Suppose that a bounded linear operator $S : X \to Y$, between real Banach spaces, must be approximated on the closed unit ball $B_X$ from $n$ continuous linear measurements. A nonadaptive deterministic algorithm has the form

$$A(x) = \phi(L_1 x, \dots, L_n x), \qquad L_j \in X',$$

where the functionals are fixed and the reconstruction $\phi : \mathbb{R}^n \to Y$ is unrestricted. Its minimal worst-case norm error is denoted by $e_n^{\mathrm{det}}(S)$. Using fewer than $n$ measurements is allowed.

A nonadaptive randomized algorithm chooses the whole list of functionals and its reconstruction independently of the input. For every seed $\omega$ it uses at most $n$ measurements. We require the real-valued error $(\omega, x) \mapsto \|Sx - A^\omega(x)\|_Y$ to be jointly measurable on the seed space times the Borel unit ball. The minimal error is

$$e_n^{\mathrm{ran}}(S) = \inf_A \sup_{x \in B_X} \mathbb{E}\|Sx - A^\omega(x)\|_Y.$$

Thus the input supremum is outside the expectation; the error criterion is the expected norm. All errors in this note use nonadaptive information.

For a bounded nonnegative sequence $u = (u_n)_{n \ge 1}$, set

$$\mathrm{rate}(u) = \sup\{\alpha \ge 0 : u_n = O(n^{-\alpha})\}.$$

Write $a(S) = \mathrm{rate}(e_n^{\mathrm{det}}(S))$, $b(S) = \mathrm{rate}(e_n^{\mathrm{ran}}(S))$, and

$$G = \sup_{\substack{X,Y \text{ real Banach}, \ S \in \mathcal{L}(X,Y) \\ a(S)<\infty, \ b(S)<\infty}} \big(b(S) - a(S)\big). \tag{1}$$

The operator and spaces in each rate are fixed as $n \to \infty$.

Krieg, Novak and Ullrich proved the general upper bound $G \le 1$ [1, Corollaries 14–15]. If either space is Hilbert, their bound improves to $1/2$. The general nonadaptive question remained between $1/2$ and 1; this interval is stated explicitly in [3, Section 10.2, equation (10.10)]. It is also the question posed in AIM Problem 555 [4]. Results that permit adaptive randomized measurements do not decide this comparison.

**Theorem 1.** *The supremum in* (1) *is* $G = 1$.

The finite-dimensional estimates needed for the lower construction have different dependence on dimension. A deterministic method has constant error on $\ell_p^m \to \ell_\infty^m$ when $m$ is sufficiently large relative to its rank. In contrast, a nonadaptive randomized sketch needs only logarithmic dependence on $m$ for a prescribed accuracy [2, Proposition 4.1]. A finite-dimensional example alone has infinite asymptotic rates, so it cannot be used in (1). We place such examples in a fixed infinite-dimensional operator. The $\ell_1$ norm across blocks controls the expected error for inputs occupying many blocks; reading an initial group of blocks exactly makes the remaining sketch budget affordable. The resulting gains approach one. We do not assert that the supremum is attained by one operator.

## 2 Finite-dimensional estimates

We first give a lower bound that allows arbitrary deterministic reconstruction. For a vector $z \in \mathbb{R}^m$, $\|z\|_r$ denotes its usual $\ell_r$ norm.

**Lemma 2.** *Let $q \geq 4$ be an even integer, $p = q/(q-1)$, and $0 \leq n < m$. Every deterministic algorithm using at most $n$ fixed linear measurements to approximate the identity $\ell_p^m \to \ell_\infty^m$ has error at least*

$$\frac{1 - \sqrt{n/m}}{1 + \sqrt{n}\, m^{-1/q}}. \tag{2}$$

*In particular its error is at least $1/4$ if $m \geq 4n$ and $m \geq n^{q/2}$.*

*Proof.* Let $N$ be the information matrix and $V$ its row space. Set

$$\rho = \max_{1 \leq i \leq m} \sup_{\substack{x \in \ker N \\ \|x\|_p \leq 1}} |x_i|.$$

The inputs $x$ and $-x$ in $\ker N$ have identical information. Their images have distance $2\|x\|_\infty$, so the worst error of any reconstruction is at least $\rho$.

For each standard basis vector $e_i$, we claim that

$$\sup_{\substack{x \in V^\perp \\ \|x\|_p \leq 1}} |x_i| = \operatorname{dist}_{\ell_q^m}(e_i, V). \tag{3}$$

Hölder's inequality proves the inequality from left to right. Choose a minimizer $v_i \in V$ for the distance. Such a minimizer exists by finite-dimensional compactness: a minimizing sequence is bounded and $V$ is closed. Put $r_i = e_i - v_i$. If $r_i = 0$, both sides vanish. Otherwise, differentiating the polynomial $\sum_j ((r_i)_j - t v_j)^q$ at its minimum $t = 0$, for $v \in V$, gives $\sum_j (r_i)_j^{q-1} v_j = 0$. Therefore

$$x_j = \frac{(r_i)_j^{q-1}}{\|r_i\|_q^{q-1}}$$

defines a vector in $V^\perp$ with $\|x\|_p = 1$ and

$$x_i = \langle e_i, x \rangle = \langle r_i, x \rangle = \|r_i\|_q.$$

This proves (3).

Form the matrix $B$ with rows $v_i$, and put $E = I - B$. Its residual rows have $\ell_q$ norm at most $\rho$, and $\operatorname{rank} B \leq n$. Hence

$$\operatorname{tr} B \geq m(1 - \rho), \qquad \|B\|_F \leq \sqrt{m} + m^{1-1/q}\rho,$$

where $\|\cdot\|_F$ is the Frobenius norm. Indeed, $\|z\|_2 \le m^{1/2-1/q}\|z\|_q$ for every row of $E$. Let $P_V$ be the Euclidean orthogonal projection onto $V$. Since $B = BP_V$, $P_V$ is symmetric and $\|P_V\|_F = \sqrt{\dim V}$, we have

$$|\operatorname{tr} B| = |\langle B, P_V\rangle_F| \le \sqrt{n}\,\|B\|_F.$$

Combining these inequalities and dividing by $m$ yields

$$1 - \rho \le \sqrt{n/m} + \sqrt{n}\,m^{-1/q}\rho,$$

which is (2). Under the two additional size conditions its numerator is at least $1/2$ and its denominator at most two. □

We use the following explicit randomized estimate from [2, Proposition 4.1], with its fixed preprint version and numbering. For integers $R, G$, with $R$ odd, draw independent uniform labels $H_{ri} \in \{1, \dots, G\}$ and signs $\sigma_{ri} \in \{-1, 1\}$. Take the $RG$ grouped linear measurements

$$y_{rg} = \sum_{\{i:H_{ri}=g\}} \sigma_{ri} z_i, \qquad C_{R,G}(z)_i = \operatorname{med}_{1\le r\le R}(\sigma_{ri} y_{r,H_{ri}}).$$

The labels and signs determine every measurement before the input.

**Proposition 3** (Kunsch–Wnuk)**.** *Let $1 \le p \le 2$, $m \ge 2$, $L \ge 1$ be an integer and $G = 2^{4+L}$. Choose $R$ as the smallest odd integer satisfying $R \ge \max\{5, 2 + 3\log_2 m\}$. Then*

$$\sup_{\|z\|_p \le 1} \mathbb{E}\|z - C_{R,G}(z)\|_\infty \le 4\,2^{-L/p}. \tag{4}$$

*The algorithm uses $RG$ measurements for every seed.*

An odd median commutes with multiplication by any real scalar. Thus (4) holds for arbitrary $z$ after multiplying its right side by $\|z\|_p$. The seed space is finite, and every realized reconstruction is continuous. These facts will also give the required measurability after the blocks are combined.

## 3 Infinite-dimensional construction

Fix an even integer $q \ge 4$, and put

$$p = \frac{q}{q-1}, \qquad s = \frac{1}{q^2}, \qquad m_k = 2^k, \qquad w_k = 2^{-sk} \quad (k \ge 1).$$

Define the real Banach spaces

$$X_q = \left(\bigoplus_{k\ge1} \ell_p^{m_k}\right)_{\ell_1}, \qquad \|x\|_{X_q} = \sum_{k\ge1} \|x_k\|_p, \qquad Y = \ell_\infty(\{(k,i) : k \ge 1,\ 1 \le i \le m_k\}),$$

and the real-linear operator

$$(S_q x)_{k,i} = w_k (x_k)_i. \tag{5}$$

The direct sum is complete: a Cauchy sequence converges in each block; passing to the limit in finite partial sums shows that its coordinate limit belongs to $X_q$ and that convergence holds in the summed norm. The same coordinate-limit argument proves completeness of $Y$. Moreover $\|S_q x\|_Y \le \|x\|_{X_q}$.

**Theorem 4.** *For the fixed operator (5), both rates are finite and*

$$s \le a(S_q) \le \frac{sq}{2}, \qquad s + \frac{1}{p} \le b(S_q) \le a(S_q) + 1. \tag{6}$$

*In particular,*

$$b(S_q) - a(S_q) \ge 1 - \frac{3}{2q} + \frac{1}{q^2}. \tag{7}$$

*Proof.* For the deterministic lower bound, let $n \ge 4$ and choose $k = \lceil (q/2)\log_2 n \rceil$. Restrict an arbitrary $n$-measurement algorithm on $X_q$ to inputs in block $k$, and project its output to that block. Its restricted information has rank at most $n$. Since $m_k \ge n^{q/2} \ge 4n$, Lemma 2 gives

$$e_n^{\mathrm{det}}(S_q) \ge \frac{w_k}{4} \ge \frac{2^{-s}}{4}\, n^{-sq/2}. \tag{8}$$

Thus $a(S_q) \le sq/2$. Conversely, reading all coordinates in blocks $1, \dots, J$ costs $2^{J+1} - 2$ measurements and leaves error $w_{J+1}$. Taking $J = \lfloor \log_2(n+2) \rfloor - 1$, for $n \ge 2$, gives $e_n^{\mathrm{det}}(S_q) \le 2^s (n+2)^{-s}$, hence $a(S_q) \ge s$.

We now allocate a strict randomized budget. In a block of dimension $m_k = 2^k$, suppose $w_k \ge \varepsilon > 0$. Apply Proposition 3 with

$$L_k = \left\lceil p \log_2 \frac{4w_k}{\varepsilon} \right\rceil.$$

Then $L_k \ge 1$, the weighted expected error is at most $\varepsilon \|x_k\|_p$, and its number of measurements satisfies

$$R_k G_k \le 6k \cdot 32 \left(\frac{4w_k}{\varepsilon}\right)^p \le C_0 k \left(\frac{w_k}{\varepsilon}\right)^p, \qquad C_0 = 3072. \tag{9}$$

Here $R_k \le 3 + 3k \le 6k$ and $1 < p \le 2$. All group measurements extend to continuous linear functionals on $X_q$, since each has support in one finite block.

Put $\theta = 2^{-sp}$, so $0 < \theta < 1$, and fix $D > 0$ such that

$$D^p \ge \frac{2C_0\, 8^{sp}}{(\ln 2)(1-\theta)^2}. \tag{10}$$

For $n \ge 8$, set

$$J = \lfloor \log_2(n/4) \rfloor, \qquad \varepsilon = D\, n^{-s-1/p} (\ln(n+1))^{1/p}, \qquad K = \max\left\{ J, \left\lfloor \frac{\log_2(1/\varepsilon)}{s} \right\rfloor \right\}.$$

Read blocks $k \le J$ exactly, sketch blocks $J < k \le K$, and return zero on the remaining blocks. These choices depend only on $n, q$. Every sketched block satisfies $w_k \ge \varepsilon$, while $w_{K+1} < \varepsilon$; if $K = J$, there are no sketched blocks.

The exact prefix costs at most $n/2$. For the other measurements, the geometric first-moment sum gives

$$\sum_{k>J} k\theta^k = \theta^{J+1}\left(\frac{J+1}{1-\theta} + \frac{\theta}{(1-\theta)^2}\right) \le \frac{(J+1)\theta^J}{(1-\theta)^2}.$$

Using $2^J \ge n/8$ and $J + 1 \le \log_2 n$, (9) yields

$$\begin{aligned} \sum_{k=J+1}^{K} R_k G_k &\le C_0 \varepsilon^{-p} \sum_{k>J} k 2^{-spk} \\ &\le \frac{C_0\, 8^{sp}}{D^p (\ln 2)(1-\theta)^2} \frac{\ln n}{\ln(n+1)}\, n \le \frac{n}{2}. \end{aligned}$$

Thus every seed uses at most $n$ measurements. There are finitely many sketched blocks, with a finite product seed space, and all their grouped measurements and reconstructions are fixed before input.

For any $x \in X_q$, bound the maximum block error by the sum of the finitely many sketched errors and the deterministic tail bound. Homogeneity of the block algorithms gives

$$\begin{aligned}\mathbb{E}\|S_q x - A_n^\omega(x)\|_Y &\le \varepsilon \sum_{J<k\le K} \|x_k\|_p + \varepsilon \sum_{k>K} \|x_k\|_p \\ &\le \varepsilon \|x\|_{X_q}.\end{aligned}$$

This includes inputs spread over infinitely many blocks. Each realized algorithm is continuous, because its output consists of finitely many linear coordinates and medians. Its scalar norm error is continuous in $x$, and hence jointly measurable on the finite seed space times $B_{X_q}$. We have proved

$$e_n^{\mathrm{ran}}(S_q) \le D\, n^{-s-1/p} \bigl(\ln(n+1)\bigr)^{1/p}, \qquad n \ge 8. \tag{11}$$

For every $\alpha < s + 1/p$, the logarithmic factor is $O(n^{s+1/p-\alpha})$, so $b(S_q) \ge s + 1/p$.

It remains to check the finite-rate restriction. The symmetric case of [1, Corollary 14] gives, for every bounded linear $S : X \to Y$ and $\alpha > 0$,

$$e_{2n-1}^{\det}(S) \le C_\alpha n^{1-\alpha} \sup_{0\le j<n} (j+1)^\alpha e_j^{\mathrm{ran,ad}}(S), \tag{12}$$

where $e_j^{\mathrm{ran,ad}}$ allows adaptive randomized information. Its continuous-information formulation is explicit in [1, Section 6.3]. Since $e_j^{\mathrm{ran,ad}}(S) \le e_j^{\mathrm{ran}}(S)$, an available nonadaptive randomized exponent $\alpha > 1$ implies $a(S) \ge \alpha - 1$. The $j = 0$ initial error is bounded by $\|S\|$, and monotonicity in the deterministic budget extends the estimate to all indices. If $b(S_q) = \infty$, choosing an available $\alpha > 1 + sq/2$ contradicts (8). Hence $b(S_q) < \infty$. If $b(S_q) > 1$, taking $\alpha$ up to $b(S_q)$ proves $b(S_q) \le a(S_q) + 1$; if $b(S_q) \le 1$, this follows from $a(S_q) \ge 0$. Finally (6) gives

$$b(S_q) - a(S_q) \ge s + \frac{1}{p} - \frac{sq}{2} = 1 - \frac{3}{2q} + \frac{1}{q^2}.$$

□

*Proof of Theorem 1.* The comparison (12) gives $b(S) - a(S) \le 1$ whenever both rates are finite. Indeed, if $b(S) > 1$, use each available $1 < \alpha < b(S)$ and then let $\alpha \uparrow b(S)$; if $b(S) \le 1$, use $a(S) \ge 0$. Thus $G \le 1$. The operators of Theorem 4 are admissible in (1) for every finite even $q \ge 4$, and their gains in (7) tend to one as $q \to \infty$. Therefore $G \ge 1$. □

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## References


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